Chapter 1

The Phonology of Nandome Dagaare

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Dagaare (Mabia) is an under-documented Niger-Congo Language spoken in Ghana and Burkina Faso by around one million people (Ali et al. 2021). This paper focuses on the Nandome dialect which has neither been described nor analyzed in the literature until now. A team of researchers centered at Georgetown University worked on a documentation project for this dialect. The present research is concerned with describing the phonological system of Nandome Dagaare. Nandome Dagaare has 23 contrastive consonants and 9 contrastive vowels (four +/-ATR pairs and a lone low vowel) which also contrast in length and nasality. The dialect has both progressive and regressive vowel harmony depending on the domain. There are two level tones (high and low), and two contour tones (rising and falling). Three allophonic alternations are also reported for Nandome Dagaare: [b]~[β], [g]~[ɣ], and [r]~[ɾ].

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1 Introduction and Language Background

Dagaare (ISO: dgi) is an under-documented Niger-Congo language of the Mabia [1] family. It is spoken in Ghana and Burkina Faso by around one million people (Ali et al. 2021). This paper is focused on the Nandome dialect, a sub-dialect of Northern Dagaare (Angsongna & Akinbo 2022; Bodomo 1997), which has neither been described nor analyzed in the literature until now. Beginning in the spring of 2022, a team of researchers at Georgetown University has been working on documenting the Nandome dialect[2] of Dagaare (ND). The present paper is concerned with describing the phonological system of this dialect.

ND, like most languages of West Africa, is tonal (§6), has Advanced Tongue Root (ATR) Vowel Harmony (VH) (§4), and its phonological processes are sensitive to particular domains (§4.2). The phonological system of ND is largely similar to that of the other dialects described in the literature (Ali et al. 2021; Bodomo 1997; Mwinlaaru & Yap 2017). There are a few differences which we highlight below; for example, ND has a nine vowel system, whereas the Sombo dialect of Central Dagaare has 10 (Ozburn et al. 2018).

The purpose of this paper is to outline the phonological system of this language. This paper serves as a preliminary account for a phonological grammar of ND. The reader should note that the data in this paper comes from elicitations conducted at Georgetown University over the course of two semesters. Furthermore, the research team had access to only one speaker and as such the data presented here cannot necessarily be considered to be representative of the whole speech community at large. We are also working with a limited data set so some of the generalizations have been drawn based on only a few tokens. That said, we believe that this phonological account is as accurate as possible at this stage.

This paper is structured as follows, in section 2 we discuss the consonants of ND. In section 3 we cover the vowels. In section 4 prosodic domains for ATR harmony processes are discussed. The syllable structure of the language is covered in 5. In section 6, we discuss tone in the language. In section 7, we cover the major phonological alternations observed in ND. Finally, section 8 concludes.

[1] Mabia was previously known as Gur.
[2] Nadome Dagaare is the term preferred by our consultant.

2 Consonants

ND has 23 consonantal phonemes and three major allophones. The phonemes are represented in the table in (1); [β], [ɣ], and [ɾ] are allophones of /b/, /g/, and /r/ respectively.

(1) Dagaare Consonant Chart

	Bilabial	Labio-dental	Dental	Alveolar	Post-alveolar	Palatal	Velar	Labio-velar
Plosive	p b		t̪ d̪				k g	k͡p g͡b
Nasal	m			n		ɲ	ŋ	ŋ͡m
Trill				r				
Tap/Flap				(ɾ)				
Fricative	(β)	f v		s z			(ɣ)	
Affricate					t͡ʃ			
Approximant						j		w
Lateral				l				

In (2) to (7) below, minimal or near minimal pairs motivate the phonemic status of many of the consonants.

(2) Stops (Voiceless vs Voiced)

a. Coronals

t̪ìɛ́	**d̪**ìɛ̀n
'think'	'play'

b. Velars

kɛ́	**g**àdò
REL ('that')	'bed'
k͡pɛ̋ɛ̋	g͡bɛ́ɛ́
'big'	'legs'

c. Labials

pàn	**b**áá
'door'	'dog'
k͡páɾó	g͡bááré
'shirt'	'finish'

(3) Fricatives (Voiceless vs Voiced)

a. Coronals

sàà	**z**àà
'rain'	'bad'

b. Labiodentals

fʊ̀	**v**ùùmí
2.SG	'fires'

(4) Nasals

a.

núú	**ɲúúr**
'hand'	'yam'

b.

gán	**báŋ**
'book'	'never'

c.

màà	**nàà**	ŋ͡màà
1SG (FOC)	'chief'	'cut'

(5) Affricates

t͡ʃén	t̪éíŋ
'go'	'floor'

(6) Liquids

píɛ́**l**	tíɛ́**r**
'near'	'think'

(7) Approximants

jír	**w**íè
'house'	'farm'

As can be seen from the tables above, phonemic distinctions are made at the bilabial, labiodental, alveolar, post-alveolar, palatal, velar, labiovelar, and glottal places of articulation. Plosives contrast at the bilabial, alveolar, velar, and labiovelar places of articulation while fricatives contrast at only the labiodental and alveolar places of articulation. There are voiced fricatives at the bilabial and velar places, but these appear only as allophones of their corresponding voiced stops. These are discussed further in section 7. Voicing is contrastive for both plosives and fricatives. Additionally, as shown in (4), nasals are contrastive at bilabial, alveolar, palatal, velar, and labiovelar places of articulation. There is only one affricate, which is voiceless and post-alveolar. There are two approximants, a palatal and a labioveolar. As seen in (6), there are two liquids, [l] and [r] which are contrastive, the latter of which has an allophone [ɾ] which will be discussed further in section 7.3.

3 Vowels

ND has a nine vowel system (Figure 1). This is in contrast to the dialect(s) described by Ozburn et al. (2018), which have been said to have a 10 vowel system. The difference is in whether the low vowel /a/ has an ATR distinction like the mid and high vowels. In the other dialects, [a] is -ATR and [ʌ] is its +ATR counterpart.[3] However, a phonetic analysis of the ND vowel inventory shows that ATR is not distinctive for the low vowel /a/ (§4.1). (8-10) below provide evidence for the phonemic status of the vowels given in Figure 1 through minimal or near minimal pairs.

[3]We set aside any discussion on the true ATR specification for the [ʌ] vowel.

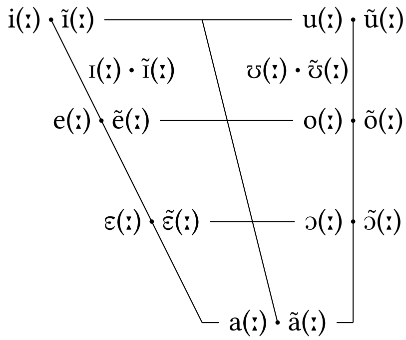

Figure 1: Nandome Dagaare Vowel Chart

(8) Height Contrasts

a. Front -ATR

tìr	tɛ̀r
'send'	'have'

b. Back +ATR

búr	bór
'get wet'	'goats'

(9) Backness Contrasts

a. High +ATR

níí	núú
'cows'	'hand'

b. Mid -ATR

dɛ̀ɛ̀	dɔ́ɔ́
'stairs'	'man'

(10) ATR Contrasts

a. Mid Front

g͡bɛ́r	g͡bèr
'leg'	'boat'

b. Mid Back

sòr	sɔ́r
'roads'	'read'

c. High Back

zú	zʊ́m
'head'	'fish' (n.)

Another property of ND vowels is that there is a nasality and length distinction. As is seen in (11), the difference between the word for 'every' and the word for 'good' is that the former has an short vowel and the latter has a long vowel. Similarly, (12) shows the contrastive difference in nasality. The word for 'rain' has an oral vowel and 'father' has a nasal vowel.

(11) Length Contrast

zà	zàà
'every'	'good'

(12) Nasality Contrast

sàà	sà̃à̃
'rain'	'father'

Therefore, our findings indicate that ND has a nine vowel system, makes both length and nasality contrasts, and demonstrates ATR vowel harmony. These findings are consistent with what has been described for other dialects of Dagaare as discussed in Bodomo (1997) but not in Ozburn et al. (2018).

4 ATR Harmony

ND has a productive ATR Vowel Harmony (VH) system, as was briefly mentioned above. As is common in ATR VH languages, words in ND either contain +ATR vowels or -ATR vowels. Furthermore, this ATR distinction is contrastive. Take (13), for example. In (13a) 'legs' has the -ATR vowel [ɛ], whereas (13b) 'boats' has the +ATR vowel [e]. When they receive the same front mid vowel as a plural suffix, it undergoes ATR harmony becoming [ɛ] in (13a) or [e] in (13b).

(13) a. g͡bɛ́-ɛ́ [-ATR]
leg-PL
'legs'

b. g͡bè-é [+ATR]
boat-PL
'boats'

As will be discussed in the below, /a/ does not participate in ATR VH. Moreover, /a/ does not block VH. This means that in a +ATR word (or -ATR one) if there are three syllables, the second of which having /a/ as a nucleus, the other two syllables will still be +ATR (or -ATR) even though they are not adjacent to each other. This can be seen in (14a) where [i] and [e] are +ATR even though /a/ intervenes between them. Likewise, in (14b) [ɪ] and [ɛ] are both -ATR despite being separted by /a/. Thus /a/ is transparent to ATR VH.

(14) /a/ Transparent to ATR VH

a. pìpá-mé
rock-PL
'rocks'

b. à-ɲíɾá-nɛ̀
DEF-ant-DEM.THAT
'those ants'

In addition to vowels in words sharing an ATR specification, any additional affix added to the word triggers harmony between the word and the affix (the directionality of ATR spread is discussed below in section 4.2). For example, the demonstrative suffix -nɛ̀ can either be +ATR or -ATR depending on the root noun it combines with. In (15a) we see that when the root noun is +ATR the demonstrative suffix is pronounced as -nè, but in (15b) when the noun root is -ATR it is pronounced as -nɛ̀.

(15) Affix Harmonization

a. à-bíè-nè
DEF-child-DEM.THAT
'that boy'

b. à-tíɛ̀-nɛ̀
DEF-tree-DEM.THAT
'that tree'

4.1 /a/

With respect to ATR distinctions for the low vowel /a/, Ozburn et al. (2018) claim that the Sombo variety of Central Dagaare has a distinction between [a] in -ATR contexts and [ʌ] in +ATR contexts. However, the accounts given by Mwinlaaru & Yap (2017) and Ali et al. (2021) say that in the Sombo dialect of Dagaare /a/ does not participate in ATR VH. A phonetic analysis of the the ND vowel inventory described below converges with the accounts given by Ali et al. (2021), Bodomo (1997), and Mwinlaaru & Yap (2017). The phoneme /a/ does not participate in ATR VH. This is in contrast to Ozburn et al. (2018) conclusion for the Sombo dialect.

For the phonetic analysis, F1 and F2 values were collected for tokens of each vowel. To ensure that there was no influence on the formant values from the surrounding consonants, all values were taken from vowels in open syllables.[4] The formant values were taken from the middle of the vowel spectrum. All measures were taken and analyzed in Praat (Boersma & Weenink 2023). The underlying assumption was that /a/ does participate in ATR VH (/a/ was considered to be +ATR if the other syllables of the prosodic word were +ATR and it was considered -ATR when the other syllables of the prosodic word were -ATR). 10 tokens were taken for each vowel (with the exception of /u/ for which only 5 tokens were available for analysis)–yielding 95 tokens in total. Next the difference in F1 values for each ATR pair was determined (F1[e] - F1[ɛ], F1[o] - F1[ɔ], etc.). This tells us the difference in height between the +ATR and -ATR variants of each vowel. On average the difference between the F1 values for ATR pairs

[4] Syllables with preceding nasals, bilabials, or velars were excluded so as to ensure that the preceding consonant has minimal effect on the following vowel's formant values.

is 65Hz (visualized in Figure 2). In contrast /a/ had no significant difference in F1 for +ATR and -ATR contexts. The interpretation of this pattern is that +ATR vowels are slightly higher than their -ATR counterparts.

Next, we compared the F2 - F1 values for +ATR and -ATR vowel pairs. We used the F2-F1 transform because it is inversely correlated with vowel backness (Zsiga 2013). Just as in the F1 analysis, we took the (F2 - F1) of the +ATR vowel and subtracted the (F2 - F1) of the -ATR vowel ((F2 - F1)[e] - (F2-F1)[ɛ], etc.). On average, the difference between these values for each ATR pair was 200 Hz (also depicted in Figure 2). The interpretation of this result is that +ATR vowels are more fronted than their -ATR counterparts. Figure 2 gives a vowel plot for the average F1 and F2-F1 values of every vowel in ND. The circle surrounds the two /a/ vowel values for expository ease.

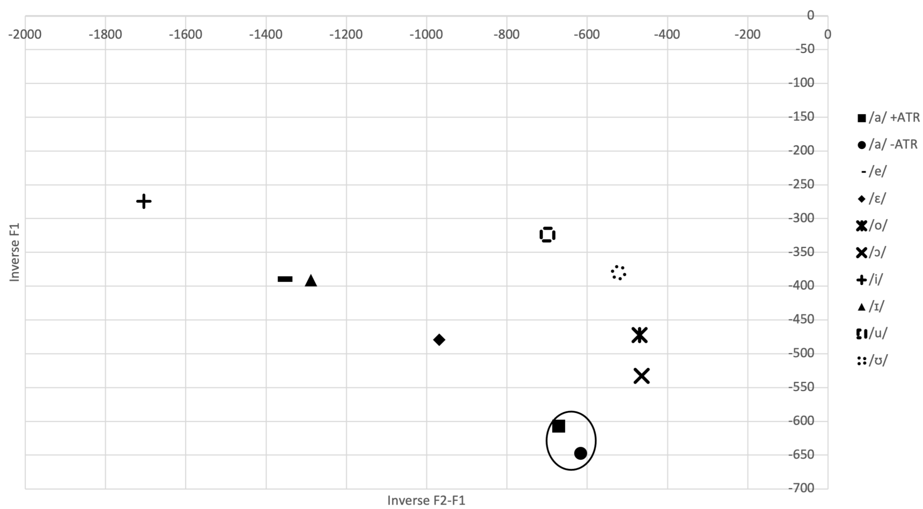

Figure 2: Nandome Dagaare Vowel Space

In addition to the overall analysis performed on all vowels, a separate analysis was conducted only for the vowel /a/. To thoroughly see the distribution of /a/ values in +ATR and -ATR contexts, 30 tokens were taken for +ATR /a/ and -ATR /a/ each. The same controlling conditions as above were made for these tokens as well. We then plotted all 60 of these points to see the overlap in their distributions. Given the high degree of overlap between the +ATR /a/ and -ATR /a/ in this vowel plot (Figure 3), for the purposes of this description we assume this is indicative of the fact that /a/ in +ATR contexts and in -ATR contexts are not phonologically distinct.

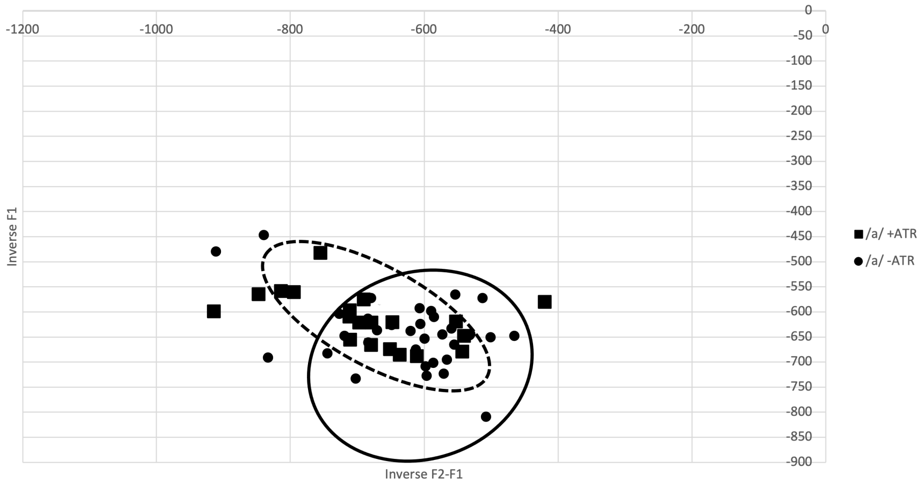

Figure 3: Distribution of /a/ tokens

This high degree of phonetic overlap suggests that there might not be any ATR distinction for the low vowel /a/. This is in line with the other dialects described by Mwinlaaru & Yap (2017) and Ali et al. (2021) but contrary to the phonetic analysis performed by Ozburn et al. (2018) on Sombo.

4.2 Domains

ATR VH in ND is both progressive and regressive. That said, the directionality of harmony is not bidirectional. Rather, progressive harmony occurs in one domain–namely the Determiner Phrase (DP), whereas regressive harmony occurs in a separate domain–the Verb Phrase (VP).

Turning first towards the verbal domain, pronouns are clitics which are underlyingly -ATR and form a prosodic constituent with the verb. If the pronoun is functioning as a subject, it occurs before the verb root, and thus undergoes regressive ATR harmony with a +ATR verb (16a). If the pronoun is functioning as an object, it occurs after the verb root and does not undergo vowel harmony (16b).

(16) Verbal Domain Harmony

a. **fù**-dé-ná à-dàà kʊ́-mì
2ND.SG-take.PRFV-AFF DEF-stick give-1ST.SG.OBJ
'you gave the stick to me'

b. ì̀-wʊ̀l-é-**fʊ́**-nì bʊ̀dɾì
1ST.SG.SUBJ-show-PROG-2ND.SG-AFF food
'I show you food'

On the other hand, in the DP, ATR VH is progressive. For example, when a +ATR noun takes a modifying adjective or a demonstrative, the adjective and demonstrative are also +ATR. Similarly, if the noun is -ATR, the following modifier will be -ATR (17)[5].

(17) Nominal Domain Harmony

a. à-dɔ̀ɔ̀-nɛ̀
DEF-man-DEM.THAT
'that man'

b. à-bíè-nè
DEF-child-DEM.THAT
'that child'

c. à-dɔ̀ɔ̀ blɛ̀
DEF-man young
'the young man'

d. à-bíè blè
DEF-child young
'the young child'

To sum up this section, our findings indicate that ND has both progressive and regressive ATR VH. Furthermore, it was established that the low vowel /a/ does not participate in the VH system. Lastly, evidence suggests that ATR VVH in ND is restricted to definable domains like the VP and/or the DP.

5 Syllable Structure

In this section, we briefly discuss the syllable structure of ND. ND syllables can have either one or two morae. Monomoraic syllables are of the form (C)V. Bimoraic syllables can be either of the following syllable structures (C)VV(C) or (C)VC(C). Onsets are optional in this dialect; as are codas. Bimoraic syllables are

[5] Note that we call this progressive harmony to distinguish it from the clearly regressive harmony seen in the verbal domain; however, this data is also consistent with root-affix harmony. There is no way to distinguish these two possibilities as the only modifier which ever occurs before nouns is the definite article /à/, which as was shown in 4.1, is transparent to VH.

syllables that have a long vowel, a diphthong, or a coda. Therefore, the following syllable types are attested in our data (18).

(18) Dagaare Syllable Types

zà	CV	'every'
ìr	VC	'raise'
lɔ̀b	CVC	'throw'
dɔ̀ɔ̀	CVV	'man'
dáár	CVVC	'sticks'

The syllable types in (18) all show up as monosyllabic roots. V and VV syllables are also attested but so far only in polysyllabic words, such as (19).

(19) a. à.ká
V.CV
'here'

b. ɔ̀ɔ̀.βá
VV.CV
'chew'

Additionally, there is one word with an unambiguous CVCC syllable (20). Whether this is simply a rare syllable type, or if this is a loan word allowed to break standard syllable restrictions is not yet clear.

(20) tʊ́βr
CVCC
'ear'

Finally, there are apparent underlying CCV morphemes that obligatorily introduce an epenthetic vowel between the two consonants in certain environments. We suspect that this is a result of a constraint against complex onsets. The reason for analyzing these words in this way comes from examples where a preceding vowel allows the first consonant of the cluster to resyllabify as the coda of the previous syllable, and thus no epenthetic vowel is needed. For example, in (20), the suffixal modifier 'young' in (20a) has no epenthetic vowel when the stem ends in a vowel and the complex onset can resyllabify with the previous syllable. However, in (20b), when the stem ends in a consonant, the complex onset cannot resyllabify and an epenthetic vowel appears. That said, we only have a handful of examples in our data set of words that behave this way so more data and further analysis is needed to determine for certain how this alternation works.

(21) Underlying CCV Syllables

a. sàà-blè ⟶ sààb.lè - 'father's younger brother'

   i. sàà - father

   ii. blè - young

b. bàpɛ́r-blɛ̀ ⟶ bà.pɛ́r.bì.lɛ̀ - 'young male friend'

   i. bàpɛ́r - male friend

   ii. blè - young

In this section, we report our findings with respect to ND syllable structure. In general, the language is fairly permissive in the types of syllables that are allowed to surface, but complex onsets and complex codas are largely dispreferred (if not banned entirely). Interestingly, our data suggests that ND has at least one syllable type not found in the other dialects described in the literature (Bodomo 1997). As far as we can tell, the VC syllable type is an innovation in this dialect that has not been described elsewhere.

6 Tone

ND makes phonemic tone distinctions with two level tones (high and low) and two contour tones (rising and falling).

(22)

a. bàà
'dog'

b. báá
'river'

c. tíɛ̀
'tree'

d. tìɛ́
'think'

Some morphological components appear to be associated with a specific tone or tonal pattern. For example, the definite prefix and demonstrative suffixes usually bear low tone, as well as pronouns, the imperfective marker, and the interrogative marker:

(23) Low-tone morphemes

a. à-wáàb-nɛ̀
DEF-snake-DEM.THAT
'that snake'

b. fù-dé-ná à-dàà kʊ́-mì
2ND.SG-take.PRFV-AFF DEF-stick give-1ST.SG.OBJ
'you gave the stick to me'

c. à-báá tì díí-rà à-ɲȁ̋ɲò
DEF-dog IMPRFV chase-PROG DEF-cat
'the dog was chasing the cat'

d. à-bìé dé-nà à-bʊ̋drì ì
DEF-child take.PRFV-AFF DEF-food Q
'did the child take the food?'

Conversely, some morphemes are usually realized with a high tone, including the habitual marker, the negative suffix, and the copula:

(24) High-tone morphemes

a. ʊ̏-mí-zɔ̀-nà-à
3RD.SG-HAB-run-AFF-Q
'does he/she run?'

b. à-báá bé-díí-rà à-ɲȁ̋ɲò jì
DEF-dog NEG-chase-IMPRFV DEF-cat NEG.POL
'the dog does not chase the cat'

c. ù-bé-nà àβé
3RD.SG-COP-AFF there
'he/she is over there'

In contrast, while the affirmative marker seems to bear a low tone, it is the most highly variable morpheme. Whether this is the effect of the Obligatory Contour Principle (OCP) or a grammatical process, we cannot say at this moment.

Beyond agglutinating morphemes, tonal patterns also serve to mark number distinctions in nouns. For example, demonstratives, nouns, and pronouns all have a high tone on the last syllable of the word in the plural. Consider the following examples (24).

(25) H tone in PL

a. dàà
stick
'stick'

b. à-dàà-nà
DEF-stick-DEM.THIS
'this stick'

c. ʊ̀
3RD.SG
'he/she/it'

d. dáá-r
stick-PL
'sticks'

e. à-dáá-r-ná
DEF-stick-PL-DEM.THIS
'these sticks'

f. bɛ́
3RD.PL.ANIMATE
'they'

In (24a) and (24b) we see that the tone on the long vowel /aː/ in the word 'stick' changes from a low tone to a high tone when the noun changes from singular to plural. Similarly in (24c) and (24d), the demonstrative -nà 'this', changes from low to high tone in the plural. Lastly, we see in (24e) and (24f) that the 3rd person pronoun has a low tone in the singular and a high tone in the plural. This suggests that there may be some degree of grammatical tone in ND. While these few examples are not conclusive in proving that there is in fact grammatical tone, there does seem to be an obvious connection between high tone and plural marking.

7 Phonological Processes

In this section, we describe several phonological processes as well as the domains in which they occur. Dagaare has at least three distinct allophones: [β], [ɣ], and [ɾ]. Furthermore, both voiceless and voiced word-final stops are always unreleased.

7.1 Voiced Bilabials

In ND the voiced bilabial plosive /b/ has a lenited allophone [β]. This section lays out the distribution of these phones. There is a general process of intersonorant lenition with several exceptions which will be explained below. First, no root-initial bilabials undergo lenition, as shown in (26):

(26) root-initial [b] in word-initial position

a. bàà
'river'

b. blè
'young'

c. bʊ̋ʊ̋
'what'

d. bàpɛ́r
'male friend'

e. bèémbè
'there'

f. bíè
'child'

g. bɔ̏ŋgánà
'bed'

h. bùsùùɾí
'clothes'

Root-initial stops also occur in multimorphemic environments (27). Regardless of the surrounding phonological environment, root-initial [b] does not alternate. In (27a-c), the definite article prefix does not cause lenition. In the compound word in (26d), the stop behaves similarly, despite the morpheme boundary. Additionally, the reduplicant in (27e) also does not undergo lenition for the second bilabial.

(27) Root-initial [b] NOT in word-initial position

a. à-bʊ́dɾì
DEF-food
'the food'

b. à-bɔ̀l
DEF-ball
'the ball'

c. à-bí-k͡páàɾà
DEF-day-morning
'the morning'

d. sɑ̃́ʊ̏-bùór
time-when
'when'

e. bʊ́ɾʊ̀bʊ́ɾʊ̀
'bread'

Regarding stops that are not root-initial, lenition does occur (except in two cases, which we address shortly). Specifically, lenition applies to stops that occur between sonorants, excepting nasals. Therefore, stops will lenite between vowels, liquids, and rhotics. (28) provides a list of words with these environments.

(28) Non-root-initial position

a.	áβé	'there'
b.	pɔ̀ɔ̀βɛ̀	'women'
c.	ɔ̀ɔ̀βʊ́	'pain'
d.	jɛ́ɛ́βɛ̀	'brothers'
e.	kʊ̀sɛ̀βɛ̀	'stones'
f.	dèβór	'when'
g.	déβlé	'boy'
h.	tʊ́βr	'ear'
i.	ɔ́βɾé	'chewing'
j.	ɔ̀ɔ̀βá	'chew'
k.	lɔ́βà	'threw'

In (28a-f), the bilabial fricative occurs intervocalically; in (28g), the fricative occurs between a lateral and a vowel; and in (28h-i), the fricative occurs between a rhotic and a vowel. Intervocalic lenition is widely attested cross-linguistically (Hualde et al. 2011; Bérces & Honeybone 2012; Hualde & Prieto 2015; Katz & Pitzanti 2019), and in the case of ND, other non-nasal sonorants trigger the same process. Figure 4 shows the spectrogram of a sentence with both bilabial allophones, displaying the difference between root initial and non-root initial bilabials.

Lenition does not occur for bilabials that precede [+ATR,+high,-back] vowels. The presence of [i] blocks the typical inter-sonorantal lentition (29).

(29) Stops before [i]

a. bìbíír
'children'

b. g͡bébíè
'toe'

c. nóbír
'fingers'

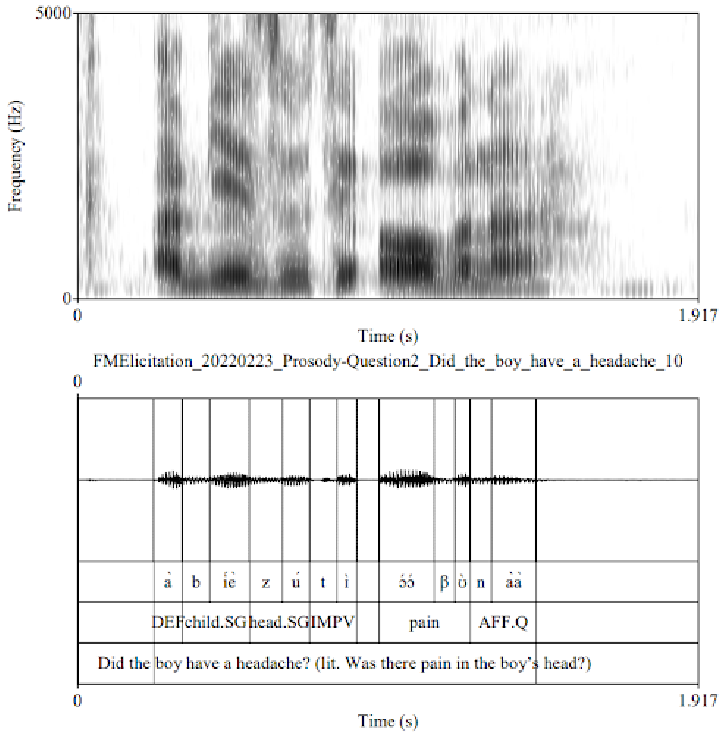

Figure 4: Spectrogram and waveform of a sentence with both bilabial allophones

Additionally, in all cases where a nasal consonants is adjacent to the bilabial, lenition does not occur (30). This is to be expected based on phonetic principles of levels of effort in production (such as Kirchner (2001)'s LAZY constraint).

(30) Nasal consonant environments

a. pàmbʊ́r
   'table'

b. jɛ́bmìnɛ́
   'brethren'

The different environments discussed thus far result in the following phonological rule for voiced bilabials which are not root intitial:

(31) /b/ ⟶ [β] / [+son,-nasal]__[+son,-nasal], except [i]

7.2 Velars

The velar alternations are quite similar to the bilabials. Underlying /g/ becomes [ɣ] between non-nasal sonorants. As with bilabials, root-initial voiced velars are immune to the lenition rule and thus surface as stops. However, we have no cases of /g/ before [i] and thus cannot establish if there is the same limitation to the lenition rule seen in the bilabials. Additionally, there is some proof of an null allomorph [Ø] in certain constructions.

(32) Velar allophones

a. gómè
 'noise'

b. bȍŋgánà
 'bed'

c. pòɣlé
 'girl'

d. à-gán
 DEF-letter
 'the letter'

We thus posit the following rule for voiced velar stops which are not root-initial:

(33) /g/ ⟶ [ɣ] / [+son,-nasal] __ [+son,-nasal]

Finally, it appears that within certain verbal constructions, the velar deletes.

(34) Deletion and vowel lengthening

a. wòɣò
 'tall'

b. à-tíɛ̀ wóómé-nà
 DEF-tree be.tall-AFF
 'the tree is tall'

Further investigation is needed to clarify the exact conditioning environment of this process as well as any potential effect of speech rate.

7.3 Rhotics

Regarding the rhotic alternation, we see a complementary distribution of the trill and the tap based on its position within the syllable. In onset position, the rhotic surfaces as a tap; and in coda position, it occurs as a trill. Syllable boundaries are shown with periods.

(35) Coda position

a. màr.bír
'star'

b. bà.pɛ́r
'male friend'

c. dáá-r
stick-PL
'sticks'

As seen in (35), the rhotic surfaces as a trill in coda position. In (36), the rhotics are in onset position and thus both surface as taps. Compare (35c) and (36c). In (336c), the plural marker resyllabifies as the onset of the vowel initial suffix and therefore becomes a tap in place of a trill. The spectrograms in Figure 5 and Figure 6 represent (35c) and (36c), respectively.

(36) Onset position

a. bʊ́d.ɾì
'food'

b. bʊ́.ɾʊ̀.bʊ́.ɾʊ̀
bread
'bread'

c. à.-dáá.-ɾ-à.tà-ná
DEF-stick-PL-3-DEM.THIS
'these three sticks'

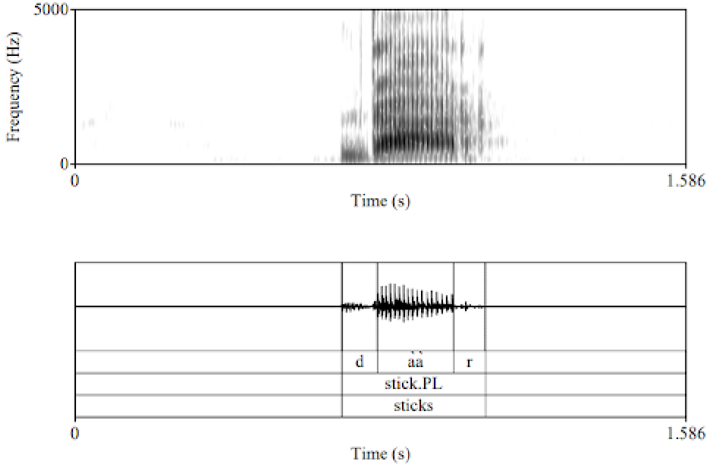

Figure 5: Spectrogram of ‘sticks’

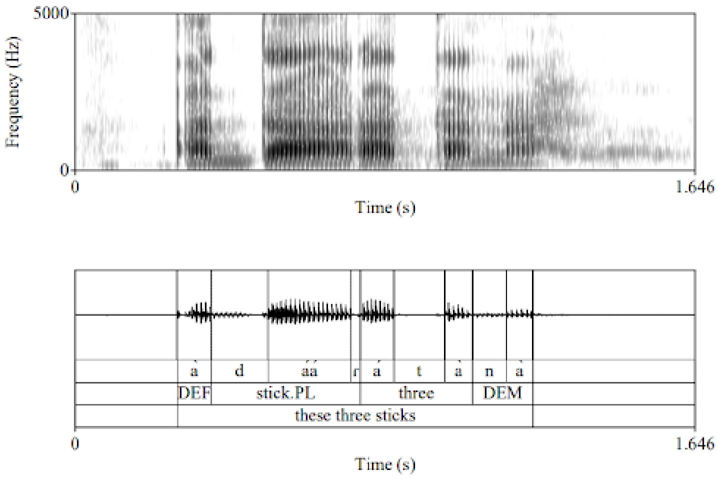

Figure 6: Spectrogram of ‘these three sticks’

The rhotics can thus be summarized by the rule in (37).[6]

(37) /r/ ⟶ [ɾ] / $[_\sigma$ __

We note that in at least one other Dagaare dialect, the trill has been described as an allophone of /d/ (Bodomo 1997). This differs from what we have seen in the ND data. We have no proof of the rhotic being an allophone of any other phonemes.

8 Conclusion

This paper provides a preliminary sketch of the phonological system of Nandome Dagaare, a dialect of Dagaare, an under-documented Mabia language spoken in Ghana and Burkina Faso. This paper functions as an introduction to the phonological system of this dialect. We provide and justify the 23 consonant phonemes and 9 vowel phonemes as well as the contrastive destinctions between short and long vowels and between nasal and oral vowels. Nandome Dagaare also has two level tones (high and low) and two contour tones (rising and falling). We show that Nandome Dagaare has progressive ATR vowel harmony in the nominal domain and regressive vowel harmony in the verbal domain. We also show that the low vowel /a/ does not participate in this process and is, in fact, transparent to it. Finally, we analyze three allophonic consonantal alternations in the langauge. We demonstrate that /b/ and /g/ becomes [β] and [ɣ] between two non-nasal sonorants so long as the second is not [i] and it is not in root initial position. Additionally, /r/ becomes [ɾ] in coda position.

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[6] Note the the decision of the underlying phoneme is essentially arbitrary and this rule could have just as easily been formulated as /ɾ/ ⟶ [r] / __ $_\sigma]$.

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